Inspired by his example, experimenters including Britain's Percy Pilcher and Americans Octave Chanute and the Wright brothers continued to explore glider flight in the last years of the century.

Yet there were serious limitations to Lilienthal's work. Although his experiments had demonstrated how essential a control system would be to any flying machine, he had failed to progress beyond control by the shifting of the pilot's body weight. Moreover, his gliders were only suitable for sport. If flight were ever to have any practical use, it would have to involve powered machines. But when Lilienthal turned his attention to powered flight, he reverted to the hopeless notion of using an engine to power flapping wings.

CHANUTE'S GLIDER EXPERIMENTS

French-born Chicagoan Octave Chanute was a wealthy middle-aged civil engineer when the "flying bug" bit him in the 1880s. A tireless communicator, he corresponded with all the aviation pioneers of his day and built up an impressive knowledge of the subject that informed his influential book *Progress in Flying Machines*, published in 1894.

Inevitably Chanute was drawn to carry out flight experiments of his own. Working with younger collaborators, notably New York aviation enthusiast Augustus Herring, Chanute developed a variety of hang gliders and, in the summer of 1896, embarked with his team to try them out among the windblown sandhills of Lake Michigan's southern shore. The site was only 30 miles (50km) from Chicago, so the activities of these oddballs soon attracted the attention of curious visitors, including reporters and news photographers.

The experimenters encountered difficulties aplenty. The winds were high and unpredictable – as soon as Chanute's group arrived in the dunes, their tent was ripped to shreds in a storm. The Lilienthal-type glider with which they began proved almost unflyable and was soon abandoned – a decision vindicated by Lilienthal's death.

Desiring a safer, more stable machine that would not require acrobatic movements of the body to fly, Chanute then tested a glider with no fewer than 12 movable wings. His principle was that "the wings should move, not the man." The multiplane glider was not, however, popular with the young collaborators who actually had to fly it.

Genuine success came late in the summer when a biplane glider designed jointly by Chanute and Herring achieved flights of up to 360ft (110m). The two wings were braced by a Pratt truss, a system that was to make an important contribution to flying-machine design. The wings were fixed and a cruciform tail assembly increased stability. The machine proved so reliable and easy to fly that visitors were invited to joyride down the dunes.

> "All agreed that the sensation of coasting on the air was delightful."
>
> **Octave Chanute**
> on the reaction of visitors who were allowed to try his gliders

DUNE FLIGHT
The Chanute-Herring biplane glides down a dune of the shores of Lake Michigan in 1896. Chanute, who was in his sixties at the time, did not attempt to fly himself.